INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location	Google, VA -- station KHEF, America/New_York
Committed pair	OSM 1252196815 -> 1287260553 Google / Google
Facade gap	207.5 m between the two halls
Weather record	42,554 real hourly records from KHEF
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.3823 C at 60 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2023-04-11 (crossing)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	1 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 3 of 24 hours with 1 mode change. The reactive incumbent took 12 hours -- 9 more -- but declared 1 unsafe hour against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 11 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	3 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	12 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s) the incumbent broke its own switch budget 1 time(s) to stay safe 11 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	5.012	18.0	5.000	0.879
01	mechanical	yes	switch budget	4.948	18.0	4.104	1.018
02	mechanical	yes	switch budget	4.118	18.0	2.434	0.914
03	mechanical	yes	switch budget	4.398	18.0	2.994	0.872
04	mechanical	yes	switch budget	3.008	18.0	1.324	0.818
05	mechanical	yes	switch budget	1.643	18.0	0.214	0.785
06	mechanical	yes	switch budget	2.143	18.0	-0.396	1.090
07	mechanical	yes	switch budget	4.673	18.0	4.104	1.396
08	mechanical	yes	switch budget	8.563	18.0	9.104	2.125
09	mechanical	yes	switch budget	11.371	18.0	12.780	2.564

10	mechanical	yes	switch budget	16.063	18.0	16.884	2.682
11	mechanical	no	dry-bulb	18.900	18.0	20.001	2.058
12	mechanical	no	dry-bulb	21.343	18.0	22.994	1.511
13	mechanical	no	dry-bulb	23.138	18.0	24.233	1.177
14	mechanical	no	dry-bulb	24.556	18.0	24.817	1.063
15	mechanical	no	dry-bulb	25.569	18.0	25.560	0.968
16	mechanical	no	dry-bulb	25.285	18.0	25.000	1.030
17	mechanical	no	dry-bulb	24.728	18.0	24.440	1.180
18	mechanical	no	dry-bulb	24.176	18.0	22.780	1.474
19	mechanical	no	dry-bulb	21.613	18.0	18.544	1.628
20	mechanical	no	dry-bulb	19.953	18.0	15.774	1.740
21	FREE	yes	-	17.453	18.0	12.434	1.589
22	FREE	yes	-	14.948	18.0	11.324	1.470
23	FREE	yes	-	12.173	18.0	8.544	1.123

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.012 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.948 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.118 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.398 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.008 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 1.643 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.143 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.673 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.563 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.372 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.063 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.900 C against a 18.0 C limit, so it fails by 0.900 C. It would take a limit 0.900 C higher, or a bound 0.900 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.343 C against a 18.0 C limit, so it fails by 3.343 C. It would take a limit 3.343 C higher, or a bound 3.343 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.138 C against a 18.0 C limit, so it fails by 5.138 C. It would take a limit 5.138 C higher, or a bound 5.138 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.556 C against a 18.0 C limit, so it fails by 6.556 C. It would take a limit 6.556 C higher, or a bound 6.556 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.569 C against a 18.0 C limit, so it fails by 7.569 C. It would take a limit 7.569 C higher, or a bound 7.569 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.285 C against a 18.0 C limit, so it fails by 7.285 C. It would take a limit 7.285 C higher, or a bound 7.285 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.728 C against a 18.0 C limit, so it fails by 6.728 C. It would take a limit 6.728 C higher, or a bound 6.728 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.176 C against a 18.0 C limit, so it fails by 6.176 C. It would take a limit 6.176 C higher, or a bound 6.176 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.613 C against a 18.0 C limit, so it fails by 3.613 C. It would take a limit 3.613 C higher, or a bound 3.613 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.953 C against a 18.0 C limit, so it fails by 1.953 C. It would take a limit 1.953 C higher, or a bound 1.953 C tighter, to change this hour.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.453 C, which is 0.547 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.589 C, and the margin is measured, not chosen: 1.575 C of group-conditional forecast error for this hour of day, plus 0.0142 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.948 C, which is 3.052 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.470 C, and the margin is measured, not chosen: 1.456 C of group-conditional forecast error for this hour of day, plus 0.0142 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.173 C, which is 5.827 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.123 C, and the margin is measured, not chosen: 1.109 C of group-conditional forecast error for this hour of day, plus 0.0142 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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